

EXERCISE 4: SPRINT PLANNING & BACKLOG GROOMING

Aim

To perform Sprint Planning, estimate user stories, manage backlog grooming, and organize development tasks using Agile methodology.

Tools Used

- Trello / Jira
- GitHub
- VS Code / Google Colab
- Planning Poker

Procedure

48. Sprint Planning Meeting

A Sprint Planning Meeting was conducted to select user stories for the sprint and define the work to be completed.

The meeting duration was 60 minutes and involved all four team members. During the meeting, the product backlog was reviewed and the following user stories were selected for the sprint:

- User Login
- Add Credit Card
- Make Payment
- OTP Verification
- Transaction History

The team discussed the scope of work and ensured that the selected stories could be completed within the sprint duration.

49. Sprint Goal Definition

The sprint goal was defined clearly to guide the development process. Sprint Goal:

To develop core functionalities including user authentication, card validation, and secure payment processing with basic security features.

50. Capacity Planning

The team capacity was calculated based on available working hours.

Team Member	Hours per Day	Number of Days	Total Hours
Member 1	5	5	25
Member 2	5	5	25
Member 3	5	5	25
Member 4	5	5	25

Total Team Capacity = 100 Hours

51. Story Point Estimation

User stories were estimated using Planning Poker technique.

User Story	Description	Story Points
US-01	User Login	3
US-02	Add Credit Card	5
US-03	Make Payment	8
US-04	OTP Verification	5
US-05	Transaction History	3

52. Task Breakdown

User stories were broken down into smaller technical tasks.

User Story: Make Payment

Task	Estimated Hours
UI Design	4
Backend Logic	6
API Integration	5
Testing	3

This breakdown helps in efficient task allocation and development.

53. Backlog Grooming

Backlog grooming was conducted to refine upcoming user stories and improve clarity.

The following activities were performed:

- Clarification of requirements
- Updating acceptance criteria
- Removing unnecessary tasks
- Improving story descriptions

54. Story Splitting

Large user stories were divided into smaller manageable tasks.

Example:

Before Splitting:

Payment System

After Splitting:

- Card Validation
- Payment Processing
- Payment Confirmation

This improves development efficiency and tracking.

55. Estimate Updates

Estimates were updated after analyzing story complexity and dependencies.

The team reviewed initial estimates and refined them to improve accuracy and reliability.



Figure 4.1: Sprint Planning Workflow

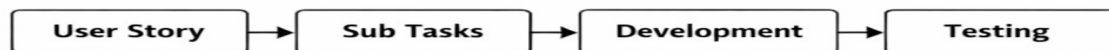


Figure 4.2: Task Breakdown Process

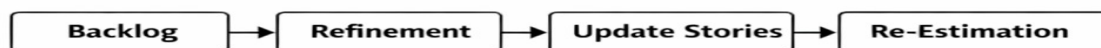


Figure 4.3: Backlog Grooming Process

Description: This diagram shows how backlog grooming improves task clarity.

Result

Successfully conducted sprint planning, story estimation, backlog grooming, and task breakdown for efficient project execution with a team of four members.